

The Jacobian is a different

An intrinsic form of Hensel's lemma: the coordinate-free invariant is $\text{Fitt}_0(\Omega_{X/\mathcal{O}})$, and $v(\det J)$ is the length of the differentials along the point

on a question of Rene Pannekoek

1. The question, and the answer in one line

Let $\mathcal{O} = \mathcal{O}_v$ be the ring of integers of K_v , complete, with uniformiser π , maximal ideal $\mathfrak{m}$ and residue field k_v . Given $F_1, \dots, F_n \in \mathcal{O}[x_1, \dots, x_n]$ and an approximate solution $a \in \mathcal{O}^n$, the multivariate Hensel lemma reads:

Hensel (Newton), coordinate version. Let $J = (\partial F_i / \partial x_j)$ and $e = v(\det J(a))$. If

$$F(a) \equiv 0 \pmod{\mathfrak{m}^{2e+1}},$$

then there is a unique $x \in \mathcal{O}^n$ with $F(x) = 0$ and $x \equiv a \pmod{\mathfrak{m}^{e+1}}$.

The question is what e is, since $\det J$ depends on the chosen equations and the chosen coordinates. The answer is that $\det J$ is not really a function but a **generator of an ideal sheaf**, and that ideal sheaf is intrinsic:

The invariant is the Kähler different

$$\mathfrak{d}_{X/\mathcal{O}} = \text{Fitt}_0(\Omega_{X/\mathcal{O}}) \subseteq \mathcal{O}_X,$$

the zeroth Fitting ideal of the sheaf of relative differentials. It is defined with no reference to coordinates; for a **square** system it is generated by $\det(\partial F_i / \partial x_j)$; and its value along a section a is

$$v(\det J(a)) = \text{length}_{\mathcal{O}}(a^* \Omega_{X/\mathcal{O}}).$$

Hensel's lemma then says: **an approximate point lifts as soon as it is a solution to precision better than the square of the different**. The failure of étaleness is the whole content of e , and $\mathfrak{d}$ is what measures it.

So the guess in the question is right — it is the sheaf of differentials — and the precise functional attached to it is the zeroth Fitting ideal. Section 2 through Section 5 make this exact. Section 6 explains why Ω **has** to be the answer, from deformation theory. Section 7 says what e measures in three familiar situations, one of which explains a constant in these notes that has never been given a reason: why square classes at $v = 2$ need a modulo 8, and at odd v only a modulo v .

2. Fitting ideals, and why they are coordinate-free

Let R be a ring and M a finitely presented R -module, with presentation

$$R^r \xrightarrow{\Phi} R^s \rightarrow M \rightarrow 0.$$

Definition. $\text{Fitt}_i(M)$ is the ideal of R generated by the $(s-i) \times (s-i)$ minors of Φ (with the conventions $\text{Fitt}_i = R$ for $i \geq s$, and the empty minor equal to 1).

Theorem (Fitting). $\text{Fitt}_i(M)$ depends only on M , not on the chosen presentation. It commutes with base change: $\text{Fitt}_i(M \otimes_R R') = \text{Fitt}_i(M)R'$.

That single sentence is the whole mechanism by which a coordinate computation becomes an invariant. Two further facts are used below.

Lemma 2.1. $\text{Ann}(M)^s \subseteq \text{Fitt}_0(M) \subseteq \text{Ann}(M)$. In particular $\text{Fitt}_0(M) = R$ if and only if $M = 0$.

Lemma 2.2. Let R be a discrete valuation ring and Φ an $s \times s$ matrix over R with $\det \Phi \neq 0$. Then

$$\text{length}_R(\text{coker } \Phi) = v(\det \Phi).$$

Proof. Smith normal form: $\Phi = UDV$ with $U, V \in \text{GL}_s(R)$ and $D = \text{diag}(\pi^{a_1}, \dots, \pi^{a_s})$. Then $\text{coker } \Phi \cong \bigoplus_i R/\pi^{a_i}$, of length $\sum_i a_i = v(\det D) = v(\det \Phi)$. ■

3. The Kähler different

Let $X \rightarrow \text{Spec } \mathcal{O}$ be locally of finite type.

Definition ([Stacks 0BVV](#), [0BVW](#)). The **Kähler different** of $X/\mathcal{O}$ is

$$\mathfrak{d}_{X/\mathcal{O}} = \text{Fitt}_0(\Omega_{X/\mathcal{O}}),$$

a quasi-coherent sheaf of ideals on X .

Three properties, all in the Stacks project's chapter on discriminants and differentials, and all exactly what is needed.

(K1) It computes the Jacobian ([0BVZ](#), Lemma 7.4). For $B = A[x_1, \dots, x_n]/(F_1, \dots, F_n)$ — n equations in n variables — the Kähler different of B over A is the ideal generated by $\det(\partial F_i/\partial x_j)$.

(K2) It cuts out the non-unramified locus ([0BVI](#), Lemma 7.3). The closed subscheme $V(\mathfrak{d})$ is exactly the set of points at which $X \rightarrow \text{Spec } \mathcal{O}$ fails to be unramified.

(K3) It commutes with base change ([0BVX](#), Lemma 7.2).

(K1) is immediate and worth seeing, because it is the entire bridge between the two formulations. For $B = A[x]/(F)$ the conormal sequence is

$$\bigoplus_{i=1}^n B F_i \xrightarrow{d} \bigoplus_{j=1}^n B dx_j \rightarrow \Omega_{B/A} \rightarrow 0, \quad dF_i = \sum_j \frac{\partial F_i}{\partial x_j} dx_j,$$

so $\Omega_{B/A} = \text{coker}(J)$ with J the Jacobian matrix, **which is square**. By the definition of Fitt_0 with $s = r = n$, the only minor is $\det J$. Hence $\mathfrak{d} = (\det J)$ — and by Fitting's theorem this ideal does not depend on the presentation, though the matrix certainly does.

4. The invariant along a point

Let $a \in X(\mathcal{O})$, i.e. a section $a : \text{Spec } \mathcal{O} \rightarrow X$.

Proposition 4.1. Suppose X is presented near a by n equations in n variables, and $\det J(a) \neq 0$. Then

$$a^* \mathfrak{d}_{X/\mathcal{O}} = (\det J(a)) \subseteq \mathcal{O}, \quad \text{and} \quad e(a) := v(\det J(a)) = \text{length}_{\mathcal{O}}(a^* \Omega_{X/\mathcal{O}}).$$

Both sides are independent of the presentation.

Proof. $a^*\Omega_{X/\mathcal{O}} = \Omega \otimes_B \mathcal{O} = \text{coker}(J(a) : \mathcal{O}^n \rightarrow \mathcal{O}^n)$, since forming Ω and forming cokernels both commute with base change. Fitting ideals commute with base change (K3), so $a^*\mathfrak{d} = \text{Fitt}_0(\text{coker } J(a)) = (\det J(a))$; and Lemma 2.2 converts that into a length. ■

For an **approximate** solution the same works, with one caveat about precision. If $F(a) \equiv 0 \pmod{\pi^N}$ then a defines a point

$$a_N \in X(\mathcal{O}/\pi^N),$$

and $a_N^*\mathfrak{d} = (\det J(a)) \subseteq \mathcal{O}/\pi^N$. This determines $e = v(\det J(a))$ **provided** $e < N$ — which is exactly the regime the theorem lives in, since $N \geq 2e + 1 > e$.

So the hypothesis of Hensel’s lemma is a statement about the point $a_N \in X(\mathcal{O}/\pi^N)$ and the scheme X , and about nothing else. One may compute e in whichever coordinates are convenient, or by pure thought (Section 7), and the answer is the same.

5. Hensel’s lemma, intrinsically

Theorem 5.1 (intrinsic Hensel). Let X be an $\mathcal{O}$ -scheme which near the point in question admits a presentation by n equations in n variables. Let $a_N \in X(\mathcal{O}/\pi^N)$ and suppose

$$a_N^*\mathfrak{d}_{X/\mathcal{O}} = (\pi^e) \subseteq \mathcal{O}/\pi^N, \quad N \geq 2e + 1.$$

Then the image of a_N in $X(\mathcal{O}/\pi^{e+1})$ lifts to a point of $X(\mathcal{O})$, and the lift is unique.

Proof. Choose any square presentation. By (K1) and Proposition 4.1, $\det J(a)$ generates $a_N^*\mathfrak{d}$, so $v(\det J(a)) = e$; and $a_N \in X(\mathcal{O}/\pi^N)$ says exactly $F(a) \equiv 0 \pmod{\pi^N}$, so $F(a) \equiv 0 \pmod{\pi^{2e+1}}$. The coordinate version of Section 1 applies and returns x with $F(x) = 0$ and $x \equiv a \pmod{\pi^{e+1}}$; the last congruence is the assertion that x lifts the image of a_N in $X(\mathcal{O}/\pi^{e+1})$. Uniqueness is the uniqueness in the coordinate version. ■

Two remarks on the shape of the statement, both of which the coordinate version obscures.

The conclusion cannot be “ a_N itself lifts”. The lift only agrees with a to precision $e + 1$, not to precision N . This is not a defect of the proof: with $X = \text{Spec } \mathcal{O}[x]/(\pi x)$ and $a = \pi^{N-1}$ one has $e = 1$ and N as large as one likes, but the only $\mathcal{O}$ -point is $x = 0$, which agrees with a to precision $N - 1$ and no better. The intrinsic statement has to be about the **truncation** of a_N , and $e + 1$ is the right truncation.

The case $e = 0$ is the classical statement. By (K2), $\mathfrak{d} = \mathcal{O}_X$ at a means X is unramified over $\mathcal{O}$ there; with flatness that is étale. So:

Hensel’s lemma, stripped: an étale $\mathcal{O}$ -scheme has $X(\mathcal{O}) \rightarrow X(k_v)$ bijective. The general case is a quantitative degeneration of this, and e is exactly the number of digits by which the scheme fails to be étale along the point.

6. Why it has to be Ω

The appearance of $\Omega_{X/\mathcal{O}}$ is not a coincidence of the formula for $\Omega_{B/A}$. Lifting points is a deformation problem, and deformation problems along square-zero extensions are governed by the cotangent complex, whose H^0 is Ω .

Precisely: let $A' \twoheadrightarrow A$ be a square-zero extension of $\mathcal{O}$ -algebras with kernel I , and let $x : \text{Spec } A \rightarrow X$ be an $\mathcal{O}$ -morphism. Then

- the obstruction to lifting x to $\text{Spec } A' \rightarrow X$ lies in $\text{Ext}_A^1(Lx^*L_{X/\mathcal{O}}, I)$;
- if a lift exists, the set of lifts is a torsor under $\text{Hom}_A(x^*\Omega_{X/\mathcal{O}}, I)$.

Take $A' = \mathcal{O}/\pi^{N+1} \twoheadrightarrow \mathcal{O}/\pi^N = A$, with $I \cong k_v$. Then:

- **X smooth over $\mathcal{O}$:** Ω is locally free, the obstruction vanishes, lifts always exist and form a torsor of size $\#\text{Hom}(x^*\Omega, k_v) = \#k_v^d$. This is formal smoothness.
- **X étale:** $\Omega = 0$, so there is no obstruction and **no ambiguity**: the lift is unique at every stage, and the inverse limit is a unique $\mathcal{O}$ -point. This is Hensel's lemma, and the proof is one line once the machine is set up.
- **X neither:** $x^*\Omega \neq 0$ is a torsion module of length e . It measures the ambiguity at each stage directly, and it controls the obstruction. Newton's method is what converts “ e digits of ambiguity per step” into the global threshold $2e + 1$: each Newton step doubles the precision and loses e to the inversion of J , so $N \mapsto 2N - e$, which is a contraction exactly when $N > 2e$.

That last line is the whole quantitative content. The iteration $N \mapsto 2N - e$ has $N = 2e$ as its fixed point: starting below it, precision decays; starting strictly above it, precision runs away to infinity. The threshold $2e + 1$ is not an artefact of the estimates — it is the fixed point of the Newton recursion, and Section 7 shows it is attained.

7. What e measures

7.1. Collision of branches

Take $n = 1$, F monic, and a an exact root, so $F = (x - a)G$. Then $F'(a) = G(a)$ and, factoring $G = \prod_{\beta \neq a} (x - \beta)$ over $\overline{K}_v$,

$$e = v(F'(a)) = \sum_{\beta \neq a} v(a - \beta).$$

So e is the total order of contact between the branch through a and all the other branches of X . If $e = 0$ the section a is an isolated, clopen branch — $\text{Spec } \mathcal{O}$ is a connected component of X near a — and one digit suffices to find it. If $e > 0$, other branches sit within distance π^e of a , and no approximation coarser than that can tell them apart. The threshold $2e + 1$ says: **approximate past the collision radius, with one digit to spare, and the root is pinned down.**

7.2. The different, in the arithmetic sense

The name is not an analogy. The main theorem of the Stacks chapter is that for a quasi-finite **syntomic** (flat and lci) morphism, the different — defined by the trace element on the relative dualising module — **agrees with the Kähler different**. For $\mathcal{O}_L$ over $\mathcal{O}_K$ with L/K finite separable and the residue extension separable, $\mathcal{O}_L$ is monogenic, $\mathcal{O}_L = \mathcal{O}_K[\theta]$ with minimal polynomial f , and

$$\mathfrak{d}_{\mathcal{O}_L/\mathcal{O}_K} = (f'(\theta)), \quad e = v_L(f'(\theta)) = d(L/K),$$

the classical different exponent: $d = e_{L/K} - 1$ when L/K is tamely ramified, and $e_{L/K} \leq d \leq e_{L/K} - 1 + v_L(e_{L/K})$ in the wild case. Checked in `hensel-different.gp`:

f	$N(\mathfrak{d}_{\mathcal{O}_K})$	$N(f'(\theta))$	equal?
$x^2 - 2$	8	8	yes
$x^2 + 1$	4	4	yes
$x^3 - 2$	108	108	yes
$x^2 - 5$	5	20	no

The last row is worth the space. $\mathbb{Z}[\sqrt{5}]$ is **not** the maximal order — $\mathcal{O}_K = \mathbb{Z}[(1 + \sqrt{5})/2]$, of index 2 — and $20 = 5 \cdot 2^2$ accordingly. The moral is exactly the one the intrinsic formulation demands:

$\text{Fitt}_0(\Omega)$ is an invariant of the **$\mathcal{O}$ -algebra one presents**, not of its normalisation. That is a feature: Hensel’s lemma is a statement about the scheme X cut out by the given equations, singularities and all, and e must see the singularities. If one silently normalises, the invariant drops and the theorem becomes false.

7.3. Why square classes need 8 at $v = 2$ and p at odd v

This is the case used on nearly every page of these notes, and it has an exact explanation.

Take $X = \text{Spec } \mathcal{O}[x]/(x^2 - u)$ for a unit u . One equation, one variable, and

$$\mathfrak{d}_{X/\mathcal{O}} = (2x) = (2) \quad \text{since } x \text{ is a unit .}$$

So $e = v(2)$ — **zero at every odd place and one at $v = 2$** — and Theorem 5.1 gives the threshold $2e + 1$:

place	$e = v(2)$	$2e + 1$	criterion for $u \in (\mathcal{O}^\times)^2$	$\#\mathbb{Q}_p^\times / \text{squares}$
v odd	0	1	u a square mod p	4
$v = 2$	1	3	$u \equiv 1 \pmod{8}$	8

The “mod 8” is the different. Every time these notes say “four square classes at odd v , eight at $v = 2$ ”, or “for a odd the curve over $\mathbb{Q}_2$ depends only on a modulo 8”, they are quoting $2e + 1$ for $e = v(2) = 1$. The place $v = 2$ is not awkward for a computational reason; it is awkward because **the squaring map is ramified there and nowhere else**, and e is exactly the ramification.

And the threshold is **attained**, so nothing coarser will do. `hensel-different.gp` sweeps the units: at $p = 2$, “square mod 8” agrees with “square in $\mathbb{Z}_2$ ” on every unit tested, while “square mod 4” disagrees on four of them, the first being

$$u = 5 : \quad 5 \equiv 1 \pmod{4} \quad \text{is a square mod 4,} \quad \text{but} \quad 5 \equiv 5 \pmod{8} \quad \text{is not a square in } \mathbb{Z}_2.$$

Precision $2e = 2$ fails, precision $2e + 1 = 3$ succeeds: the bound of Theorem 5.1 is sharp.

8. Beyond the square case

8.1. More variables than equations

If $X = V(F_1, \dots, F_r) \subset \mathbb{A}^n$ has relative dimension $d = n - r > 0$, “the Jacobian” is not even a square matrix, and the coordinate statement has to be phrased as “**some** $r \times r$ minor has valuation e ”. Intrinsically there is no choice to make:

The Jacobian ideal is $\mathfrak{J}_{X/\mathcal{O}} = \text{Fitt}_d(\Omega_{X/\mathcal{O}})$.

With n generators dx_j and r relations dF_i , Fitt_d is the ideal of $(n - d) \times (n - d) = r \times r$ minors of the Jacobian — so the intrinsic ideal is generated by **all** the maximal minors at once, and $e = v(a^*\mathfrak{J})$ is the minimum of their valuations, which is what “some minor” was groping at. Smoothness of relative dimension d is $\mathfrak{J} = \mathcal{O}_X$, and $\mathfrak{J} = \mathfrak{d}$ when $d = 0$.

Theorem 5.1 holds verbatim with $\mathfrak{d}$ replaced by $\mathfrak{J}$.

8.2. Non-complete-intersections: Elkik’s ideal

If X is presented by more equations than its codimension, $\text{Fitt}_d(\Omega)$ is still intrinsic but is no longer the right invariant: it does not know which subsystems cut out X and which are redundant. The repair is due to Elkik ([Ann. Sci. ÉNS 6 \(1973\) 553–603](#)). For $B = A[x_1, \dots, x_n]/I$ she defines an ideal $H_{B/A}$ generated by the products

$$\Delta_\alpha \cdot k, \quad k \in \left((F_{\alpha_1}, \dots, F_{\alpha_p}) : I \right),$$

over all subsystems α of I and all $p \times p$ minors Δ_α of the Jacobian of that subsystem — the minor is weighted by an ideal quotient measuring how far the subsystem is from generating all of I . Its vanishing locus is again the non-smooth locus, and her theorem carries the same exponents: **if $\pi^h \in H(a)$ and $F(a) \equiv 0$ modulo π^N with $N > 2h$, the approximate solution can be corrected to an exact one, congruent to a modulo π^{N-h} .**

Setting $N = 2h + 1$ recovers a lift agreeing to precision $h + 1$, which is Theorem 5.1. And in the square lci case the full subsystem has $(I : I) = B$, so $H_{B/A}$ contains the Kähler different and the classical criterion is already available.

8.3. When $e = \infty$

If a lies on the singular locus, $a^*\mathfrak{J} = 0$ and no threshold exists: Theorem 5.1 says nothing, and it should not, since points can genuinely fail to lift. What survives is Greenberg’s theorem: for X of finite type over $\mathcal{O}$ there are constants c, N_0 such that for $N \geq N_0$, every point of $X(\mathcal{O}/\pi^{cN})$ has the same image in $X(\mathcal{O}/\pi^N)$ as some genuine point of $X(\mathcal{O})$. That is the qualitative shadow of $2e + 1$ once e is allowed to be unbounded, and Elkik’s theorem is what makes it effective.

9. The dictionary

coordinate version	intrinsic version
the Jacobian matrix $J = (\partial F_i / \partial x_j)$	the presentation matrix of $\Omega_{X/\mathcal{O}}$
$\det J$	a generator of $\mathfrak{d}_{X/\mathcal{O}} = \text{Fitt}_0(\Omega_{X/\mathcal{O}})$
$\det J(a)$ is a unit	$X \rightarrow \text{Spec } \mathcal{O}$ is unramified at a
$e = v(\det J(a))$	$e = \text{length}_{\mathcal{O}}(a^*\Omega_{X/\mathcal{O}})$
$F(a) \equiv 0 \pmod{\mathfrak{m}^N}$	a is a point of $X(\mathcal{O}/\pi^N)$
$N \geq 2e + 1$	the point is a solution to better than $\mathfrak{d}^2$
$x \equiv a \pmod{\mathfrak{m}^{e+1}}$	the lift refines the image of a in $X(\mathcal{O}/\pi^{e+1})$
some $r \times r$ minor is non-zero	$\text{Fitt}_d(\Omega_{X/\mathcal{O}})$, d the relative dimension
redundant equations spoil the minors	Elkik’s $H_{B/A}$, minors weighted by $(F_\alpha : I)$

10. Where this bites in these notes

The threshold $2e + 1$ with $e = v(2)$ is the reason for a pattern that runs through the whole corpus and has so far been treated as a brute fact.

- `localimg.gp` computes local images over “4 classes for odd v , 8 classes for $v = 2$ ”; `p2.gp` is a separate script **only** because $p = 2$ needs 8 square classes. Both are Section 7.3.
- `selmer-involution.typ` proves its theorem at $v = 2$ by an exhaustive check over a modulo 8, and observes that “over $\mathbb{Q}_2$ every unit is a cube, so the cubic part of any sextic twist by a unit is trivial”. That asymmetry is the same computation for $x^3 - u$ in place of $x^2 - u$: there $\mathfrak{d} = (3x^2) = (3)$, so $e = v(3) = 0$ at $v = 2$ and the cube map is **étale** over $\mathbb{Z}_2$ — one digit suffices, and every unit is a cube. **Squares are hard at 2 and cubes are easy, for the same reason and with the same invariant.**
- `cover2.gp` asks whether every point of $X(\mathbb{Z}_p)$ modulo p^k is hit by a rational point, and has to choose k . The right k is $2e + 1$ for e the valuation of the Jacobian ideal along the relevant section, and the intrinsic formulation says that number can be computed from X rather than discovered by experiment.

Summary. The invariant is $\mathfrak{d}_{X/\mathcal{O}} = \text{Fitt}_0(\Omega_{X/\mathcal{O}})$, the Kähler different: an ideal sheaf on X , generated by $\det J$ in any square presentation, cutting out the locus where X is not unramified over $\mathcal{O}$, and equal to the classical different when X is a quasi-finite syntomic $\mathcal{O}$ -scheme. Along a section it evaluates to (π^e) with $e = \text{length}(a^*\Omega)$. Hensel's lemma says an approximate point lifts as soon as it solves the equations to precision better than $\mathfrak{d}^2$, and the lift is pinned down to precision $\mathfrak{d}$. In relative dimension d replace Fitt_0 by Fitt_d ; without lci, replace it by Elkik's $H_{B/A}$; on the singular locus, replace the theorem by Greenberg's.